Project TitleSkill:-

Bridge Career Navigator (CareerNav AI)

1. Introduction

Skill-Bridge Career Navigator is an AI-powered platform designed to help students and early-career professionals bridge the gap between their current skill set and industry job requirements. The system analyzes user profiles (resume, GitHub, experience) and generates a personalized roadmap to achieve a target role.

2. Problem Statement

Students and career switchers often struggle to:

- Identify required skills for specific roles
- Understand their readiness level
- Find structured learning paths

This results in inefficient preparation and lack of direction.

3. Target Audience

- 🎓 Recent Graduates
- 🔄 Career Switchers
- 👩‍🏫 Mentors guiding students

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4. Solution Overview

The platform:

1. Accepts user input (resume, GitHub, experience)
2. Uses AI to analyze skill gaps
3. Provides:
    - Readiness Score
    - Missing Skills
    - Personalized Learning Roadmap
    - Mock Interview Questions

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5. System Architecture

🔹 High-Level Architecture

Frontend (React)
↓
FastAPI Backend
↓
AI Layer (LLM API / Fallback Logic)
↓
Response Processing
↓
Frontend Display

🔹 Components

1. Frontend

- React + TypeScript

- Tailwind CSS
- Pages:
  - Home
  - Analyze
- Components:
  - ProfileForm
  - GapAnalysis
  - Roadmap
  - InterviewPrep

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2. Backend

- FastAPI
- Handles:
  - API requests
  - Input validation
  - AI integration
  - Fallback responses

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3. AI Layer

- Uses LLM (Anthropic / Gemini / OpenAI)
- Tasks:
  - Skill extraction
  - Gap analysis

    - Roadmap generation
    - Question generation

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4. Fallback System (IMPORTANT for evaluation ✅)

When API key is missing or fails:

- Returns **predefined structured mock data**
- Ensures full functionality without AI

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6. Data Flow

1. User submits profile
2. Request sent to /api/analyze
3. Backend:
    - Validates input
    - Calls AI OR fallback
4. Response generated:
    - Readiness score
    - Skill gaps
    - Roadmap
5. Frontend displays results

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7. Core Features

✅ **1. Gap Analysis Dashboard**

- Compares user skills vs target role
- Outputs missing skills

✅ **2. Dynamic Learning Roadmap**

- Step-by-step plan
- Prioritized learning

✅ **3. Mock Interview Generator**

- Role-specific technical questions

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8. API Design

🔹 **GET /api/health**

- Health check

🔹 **POST /api/analyze**

Input:

```
{
  "resume_text": "Resume content",
  "target_role": "Cloud Engineer",
  "github_url": "https://github.com/user",
  "name": "Alex",
  "education": "B.Tech",
  "experience": "1 year"
}
```

Output:

```
{
  "readiness_score": 70,
  "missing_skills": ["Kubernetes", "AWS"],
  "roadmap": ["Learn AWS basics", "Build projects"],
  "analysis": "You are close to being job-ready..."
}
```

- **POST /api/interview-questions**

Input:

```
{
  "resume_text": "...",
  "target_role": "Cloud Engineer",
  "skill_gaps": ["AWS", "Terraform"]
}
```

Output:

```
{
  "questions": ["Explain AWS EC2", "What is Terraform?"]
}
```

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9. Tech Stack

Layer	Technology
Frontend	React, TypeScript, Tailwind
Backend	FastAPI
AI	LLM APIs (Anthropic / Gemini)

Layer	Technology
Database	Optional (currently stateless/demo mode)

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10. Meeting Requirements :

✓ Core Flow

- End-to-end flow: Input → Analysis → Output

✓ AI Integration + Fallback

- AI used for analysis
- Fallback returns mock data

✓ Basic Quality

- Input validation implemented
- Error handling included
- Demo mode ensures stability

✓ Data Safety

- Uses synthetic data only

✓ Security

- API keys stored in .env
- .env.example provided

📌 11. Future Enhancements

- Resume parser using NLP
- Real job dataset integration

- ATS score checker
- Live mock interviews
- Skill tracking dashboard

📌 12. Conclusion

Skill-Bridge Career Navigator provides a structured, AI-driven approach to career planning, helping users clearly understand their next steps and reduce uncertainty in their career journey.